

Data Analysis Learning Roadmap

A free, self-paced curriculum with topics, subtopics, and clickable resources (YouTube, official sites, free certificate courses) to build real skills and strengthen your resume.

How to use this guide: Work through the sections in order — each builds on the last. Spend 60–90 minutes a day. Aim for 8–12 weeks total if going part-time. Every resource below is free. Click any blue link to open it.

1 Foundations of Data Analysis

Understand what analysts actually do before touching any tool.

- What a data analyst does day-to-day
- Types of data: qualitative vs quantitative, structured vs unstructured
- The analysis lifecycle: Ask → Prepare → Process → Analyze → Share → Act
- Data-driven decision making and KPIs
- [Alex The Analyst — YouTube channel \(full free roadmap\)](#)
- [Google Data Analytics Certificate \(apply for Financial Aid = free\)](#)

2 Excel / Spreadsheets

The most-used tool in real analyst jobs. Master this first.

- Core formulas: SUM, IF, COUNTIF, SUMIFS
- Lookup functions: VLOOKUP, XLOOKUP, INDEX-MATCH
- PivotTables and PivotCharts
- Data cleaning: removing duplicates, text-to-columns, data validation
- Basic dashboards and conditional formatting
- [excelisfun — YouTube \(3,700+ free Excel/Power BI videos\)](#)
- [ExcelJet — quick formula reference](#)

3 SQL (Structured Query Language)

The #1 required skill in most data analyst job postings.

- SELECT, WHERE, ORDER BY, filtering data
- Aggregations: GROUP BY, HAVING, COUNT/SUM/AVG
- JOINS: INNER, LEFT, RIGHT, FULL
- Subqueries and Common Table Expressions (CTEs)
- Window functions (RANK, ROW_NUMBER, LAG/LEAD)
- [W3Schools — interactive SQL tutorial \(try queries in-browser\)](#)
- [Luke Barousse — free SQL for Data Analytics course + certificate](#)
- [Luke Barousse — YouTube channel \(SQL, Python, Excel, Power BI\)](#)

4 Statistics & Probability Basics

You don't need a math degree, but these fundamentals separate analysts who guess from analysts who reason.

- Mean, median, mode, standard deviation
- Distributions and outliers
- Correlation vs. causation
- Basics of hypothesis testing
- Simple linear regression concepts
- [Khan Academy — free Statistics & Probability course](#)

5 Python for Data Analysis

Adds automation and scalability beyond what Excel/SQL can do.

- Python basics: variables, loops, functions
- Pandas: DataFrames, filtering, grouping, merging
- NumPy for numerical operations
- Data cleaning at scale (missing values, duplicates, type conversion)
- Matplotlib / Seaborn for visualization
- [freeCodeCamp — Data Analysis with Python \(free certificate\)](#)
- [Kaggle Learn — free Pandas, Python & Data Viz micro-courses](#)
- [Pandas official “Getting Started” guide](#)

6 Data Visualization & BI Tools

Turning analysis into dashboards and stories stakeholders can act on.

- Power BI: Power Query, data modeling, DAX basics, report building
- Tableau: connecting data, building visualizations, dashboards
- Choosing the right chart type for the message
- Dashboard design and storytelling with data
- [Microsoft Learn — free Power BI learning paths](#)
- [Tableau — official free training & tutorials](#)
- [Storytelling with Data — free articles on visual best practice](#)

7 Build a Portfolio (Most Important for Resume)

Certificates get you an interview; a portfolio gets you hired. Do 2–3 real end-to-end projects.

- Pick a public dataset (sales, sports, YouTube, COVID, finance, etc.)
- Clean and explore it (Excel/Python)
- Query it with SQL to answer specific business questions
- Visualize findings in Power BI or Tableau
- Publish the project + write-up on GitHub and LinkedIn
- [Kaggle Datasets — free public datasets for practice projects](#)
- [Alex The Analyst — full portfolio project walkthroughs](#)

8 Free Certificates Worth Adding to Your Resume

Apply for Coursera Financial Aid on Google/IBM certificates to get them for \$0.

- [Google Data Analytics Professional Certificate \(Coursera\)](#)
- [IBM Data Analyst Professional Certificate \(Coursera\)](#)
- [Kaggle Learn certificates \(instant, 100% free\)](#)
- [freeCodeCamp Data Analysis with Python certificate](#)
- [Luke Barousse SQL for Data Analytics certificate](#)
- [LinkedIn Learning: Career Essentials in Data Analysis \(Microsoft & LinkedIn\)](#)

9 Resume & LinkedIn Tips

- List tools by proficiency: Excel, SQL, Python, Power BI/Tableau
- Turn each portfolio project into a resume bullet with a measurable result (e.g. “Identified a 15% drop in Q3 retention using SQL cohort analysis”)
- Add certificates under a dedicated “Certifications” section, not buried in Education
- Pin your best 2 GitHub projects and link them directly in your LinkedIn “Featured” section
- Use the exact keywords from job postings (e.g. “dashboarding”, “stakeholder reporting”) in your summary

Suggested order: 1 Foundations → 2 Excel → 3 SQL → 4 Statistics → 5 Python → 6 Visualization → 7 Portfolio (build alongside 3–6) → 8 Certificates (apply for financial aid early, it takes ~15 days) → 9 Resume.

All resources listed are free at the time of writing. Availability of individual free courses can change — verify access when you click through.